1. Item 4: Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

- The problem provides a textual description (inner diameter 2 metres, three wings, 4 rotations per minute, maximum of 2 people per sector, 30 minutes) and a plan view diagram (not shown). One must correctly understand the geometry, motion, and capacity constraints of the revolving door. This is the starting point for modeling.

2. Item 2: Identify the key variables in a model

- Key variables include: rotation speed (4 rotations/minute), number of sectors (3), maximum people per sector (2), total time (30 minutes). These directly determine throughput.

3. Item 5: Recognize mathematical structure (including regularities, relationships, and patterns) in problems or situations

- The problem implies a periodic pattern: each full rotation brings each sector to the entrance once, carrying up to 2 people. Thus total people = rotation speed × number of sectors × people per sector × time. This is a typical multiplicative relationship.

4. Item 7: Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

- In reality, people take time to enter/exit, door wings have thickness, but the problem does not provide these details. Modeling must ignore such factors, assuming instantaneous entry/exit, constant rotation speed, no obstruction, leading to an ideal throughput model.

5. Item 9: Translate a problem into a standard mathematical representation or algorithm

- Translates into arithmetic: total people = 4 (rot/min) × 3 (sectors/rot) × 2 (people/sector) × 30 (min) = 720. This is a standard multiplication algorithm.

6. Item 1: Select a mathematical description or a representation that describes a problem

- Could use a rate model: throughput = 24 people/min, total = rate × time. Or simple integer multiplication. Choose a concise arithmetic representation.

7. Item 3: Select a representation appropriate to the problem context

- The problem involves continuous rotation and discrete numbers of people, so integer multiplication and time unit (minutes) are appropriate; no complex equations needed.

8. Item 6: Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

- Similar to item 2, but more explicitly describing the "mathematical aspects": e.g., periodicity of rotation, equal division of sectors, upper capacity limit, accumulation over a time interval.

9. Item 8: Recognize aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

- This problem resembles an "assembly line throughput" or "rotary conveyor" problem. The known solution is to calculate the flow rate per unit time and multiply by total time.

10. Item 10: Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

- For example, define rotation speed $r = 4$ rot/min, number of sectors $n = 3$, capacity $c = 2$, time $t = 30$, then total people $N = r \times n \times c \times t$. A diagram could also show rotation cycles relative to the entrance.

11. Item 12: Identify the constraints, assumptions, simplifications in a mathematical model

- Explicit assumptions: people enter/exit instantly, constant rotation without stopping, each sector always has two people waiting, ignore wing thickness and body size, no congestion at exit. These are necessary to arrive at 720.

1. 第4条：阅读、解码并理解陈述、问题、任务、对象或图像，以创建情境模型

- 题目提供了文字描述（内径2米、三个翼、每分钟4转、每扇区最多2人、30分钟）和平面示意图（未显示），需要正确理解旋转门的几何结构、运动规律及容量限制。这是建模的起点。

2. 第2条：识别模型中的关键变量

- 关键变量包括：旋转速度（4转/分钟）、扇区数量（3个）、每个扇区最大人数（2人）、总时间（30分钟）。这些变量直接决定吞吐量。

3. 第5条：识别问题中的数学结构（规律、关系、模式）

- 问题隐含周期性规律：每转一圈，每个扇区经过入口一次，可容纳2人。因此总人数 = 转速 × 扇区数 × 每扇区人数 × 时间。这是典型的乘法关系。

4. 第7条：简化或分解情境，使其适合数学分析

- 实际旋转门中，人员进出需要时间，门翼有厚度，但题目未提供这些细节。建模时需忽略这些因素，假设人员瞬间进出、门匀速旋转、无阻挡，从而简化为理想吞吐量模型。

5. 第9条：将问题转化为标准数学表示或算法

- 转化为算术运算：总人数 = 4 (转/分钟) × 3 (扇区/转) × 2 (人/扇区) × 30 (分钟) = 720。这是标准的乘法算法。

6. 第1条：选择描述问题的数学描述或表示

- 可采用速率模型：吞吐率 = 24人/分钟，总人数 = 速率 × 时间。或者用整数乘法。选择简洁的算术表示。

7. 第3条：选择适合问题情境的表示

- 问题中涉及连续旋转和离散人数，适合用整数乘法和时间单位换算（分钟）表示，无需复杂方程。

8. 第6条：识别并描述真实问题情境中的数学方面，包括重要变量

- 与第2条类似，但更强调对“数学方面”的显式描述，例如：旋转的周期性、扇区的等分性、容量的上限、时间区间内的累积。

9. 第8条：识别问题中与已知问题或数学概念、事实、程序相对应的方面

- 该问题类似于“流水线吞吐量”或“转盘式输送机”问题，已知解法是计算单位时间通过量乘以总时间。

10. 第10条：使用数学工具（变量、符号、图表）描述数学结构和/或关系

- 例如设转速为 $r = 4$ 转/分，扇区数 $n = 3$，容量 $c = 2$，时间 $t = 30$，则总人数 $N = r \times n \times c \times t$。也可用图表表示旋转周期与入口位置的关系。

11. 第12条：识别数学模型中的约束、假设、简化

- 明确假设：人员可瞬间进出、门匀速且无停顿、每个扇区独立且始终有两人等待、忽略门翼厚度和人员身体尺寸、不考虑出口拥堵。这些是得出720的关键。